

Project (Spring 2026)

CS- 1004 : OBJECT ORIENTED PROGRAMMING



SUPER VEHICLE-001 METAL SLUG



The treacherous General Morden has allied with alien forces and undead armies to conquer the world! The Peregrine Falcon Strike Force i.e. Marco, Tarma, Eri, and Fio, must deploy across varied terrains (land, air, and sea) to dismantle Morden's war machine. But survival is not guaranteed unless they can master powerful vehicles, diverse weapons, and the mysterious art of fusion.

[Gameplay can be watched here](#) for reference (do have a look before going ahead into the document to avoid any confusion)

There are several videos attached as hyperlinks to help you in the implementation of the project



PLAYABLE CHARACTERS

MARCO ROSSI

The pure infantry fighter. The benchmark for direct combat performance

Bufs

- 25% higher fire rate
- His melee attack can pierce through shielded opponents

Weaknesses

- Has 2 less grenades than base value
- No bonuses when piloting the slug

Special Power-up

- Allows Marco to fire in both directions simultaneously for a limited time (10 seconds)



TARMA

Mediocre on foot. Transformational on vehicle

Bufs

- 25% higher vehicle fire rate
- 20% higher vehicle durability
- If the vehicle is destroyed with Tarma in it, Tarma survives (the vehicle acts like a totem of undying)

Weaknesses

- 20% slower than base on foot
- 20% less HP on foot

Special Power up

- Becomes immune to all damage to him and the vehicle he is piloting for a limited time (20 seconds).



ERI KASAMOTO

The grenade system is her primary weapon

Bufs

- Carries double the grenades
- 50% bigger blast radius

Weaknesses

- No melee attack
- 20% lower fire rate

Special Power-up

- Allows her to throw 2 grenades at the cost of one for a limited time (10 seconds). (The extra one falls 2 blocks farther)



FIOLINA GERMI

The Weapon Master

Bufs

- When picking up a weapon, she gets 50% more ammo than base
- 10% higher fire rate than base

Weaknesses

- 25% less damage on melee attacks
- 2 less grenades than base

Special Power up

- Fire rate gets increased by 100% for a limited time(10 seconds) -

SUPERCHARGED

BASE STATS

The character spawns with a pistol with infinite ammo and also a knife. The player can pick up better weapons temporarily disabling the pistol.

Pistol Damage: 3 HP

Knife Damage: 2 HP

Pistol Fire-rate: 4 shots per second

Melee Range: Only upon collision

Melee Cooldown: 0.5s

Starting Grenades: 10

Grenade Damage: 20 HP

(Rest open to interpretation)

FUSION COMPANION

The player can spawn a "Fusion Companion" for **30 seconds ONCE** per level in Survival Mode and with a cooldown of **3 minutes** in Campaign Mode.

This "Fusion Companion" is an immortal combination of 2-4 of the above characters. This companion is computer controlled and follows the player and kills the enemies. The stats of the chosen characters to be fused are taken and then averaged to form the stats of the fusion. All the buffs and weaknesses are also inherited.

P.S. the sprite for this upto your own choice. **SUPRISE US**

SWITCHING BETWEEN CHARACTERS

Pressing the **Z Key** during the play will switch to the next character. Each character will have a separate inventory states (i.e. if Marco picked up a weapon, Tarma will not have access to it) . Each character will have exactly 2 lives, and once dead, the player can't switch to that character.



STATES

The characters can be in one of two states in the game
(reference to the video)

UNDEAD: 50% decrease in walking speed for 10 secs.

MUMMY: Converts to a mummy for 10 secs and only has a knife as a weapon.

Note: The game's video is going to be the biggest reference material. Try to look at it to understand the Vehicle and similar mechanics better



ENEMIES

INFANTRY

REBEL SOLDIER

- The most basic enemy.
- Has a pistol equipped.
- Has 2 HP.
- Spawns in batches of 2-4.



BAZOOKA SOLDIER

- Slow-moving fires a rocket at a steep arc and uses a rocket launcher.
- Has 2 HP.
- Spawns in batches of 1-2

SHIELDED SOLDIER

- Carries a riot shield that blocks frontal bullet hits entirely; must be hit from above, behind, or with explosives.
- Has a pistol and a shield.
- Has 5 HP.
- Spawns in batches of 1-2

GRENADE SOLDIER

- Lobs grenades towards the player in projectile motion.
- Has 2 HP.
- Spawns in batches of 1-2.

AERIAL

PARATROOPERS

- Descend from the top as either of the 4 infantry types.
- Inherits the batch sizes as well



UNDEAD



MUMMY WARRIOR

- Slowly walks towards the player.
- Transforms the player upon collision.
- Crumbles when hit and then, it is resurrected after 3 seconds.
- Only dies from explosions or fire.
- 5 HP.
- Spawns alone

ZOMBIES

- Slowly walks towards the player and transforms the player to undead upon collision. Has a pistol as well
- 5 HP.
- Spawns in batches of 3-5.



ALIENS

MARTIAN

- **2 PHASED.**
- In the Phase 1, it flies in a pod above the player and fires energy beams when on top.
- In Phase 2, when the pod explodes, the Martian is on foot and has a pistol.
- Has 3 HP



DAMAGE PROTOCOL FOR THE PLAYER

- When hit by a bullet, the player transitions to injury state for **exactly 1 second** and if in that state, the player is hit again, it transitions to the next state of criticality for 1 second and if it's hit again in that state, it dies.
- You have to showcase these states by adding a translucent red hue to the screen like in modern AAA games like GTA V and RDR 2.
- Same damage taking applies to player vehicles.
- Upon death the player respawns with the same inventory and loses a life.



VEHICLES

ENEMY VEHICLES

FLYING TARA

- Hovers at medium height.
- Moves horizontally.
- Drops a grenade when on top of the player.
- Spawns in batches of 2.



M-15A BRADLEY

- It's a long range artillery unit designed to be a stationary long range missile launcher.
- Has 7 HP.
- Launches missiles in projectile motion.
- They can move horizontally in plain regions to approach the player



ENEMY SUB

- It spawns in the aquatic biome and fires rockets towards the player.
- Has 7 HP.



PLAYER VEHICLES

METAL SLUG (SV-001)

- Useful in the plains biome.
- Fires rapid fire bullets.
- Must be driven by the characters.
- It can tilt (360 degree movement) and traverse 1 block differences making it flexible.



SLUG MARINER

- A faster alternative to swimming in the ocean in the aquatic biome.
- It has three types of missiles 3 each in ammo.
- A straight horizontal one and a straight vertical one (for enemies above). The third one is in reverse projectile motion towards the surface of the water



SLUG FLYER

- Can be flown and fires bullets at half the slug's speed
- Has 4 missiles built in.



AMPHIBIOUS SLUG

- It's a flyer and a mariner and a normal slug.
- When it touches water, it changes to a mariner and when it comes out of the water, it changes to a flyer, and when grounded it becomes the normal slug.

WEAPONS

FIRE ARMS

PISTOL

- Default weapon
- Always available
- Infinite Ammo
- **Note:** This uses BASE STATS ABOVE



HEAVY MACHINE GUN

- Fires rapidly when the key is held down.
- Double the fire rate of a normal pistol.
- One bullet deals the same damage as one bullet of the pistol i.e. 3 HP

ROCKET LAUNCHER

- Single rocket per trigger pull.
- Explosions on impact.
- 2 seconds reload between shots.
- Deals 5 HP in a blast radius if 3 blocks



FLAME SHOT

- Key held trigger. Shoots a stream of fire for 5 blocks
- The enemies caught in the stream take 2 HP damage per second.
- The only way to kill the mummies is this weapon as they die instantly from fire.



LASER GUN

- Spawns a beam till the end of the screen in the facing direction when triggered.
- Instantly kills the enemies upon impact.



GRENADES

HAND GRENADE

- The default held grenade by all the characters.
- Ballistic arc throw.
- Deals 5 HP in a blast radius of 3 blocks



FIRE BOMB GRENADE

- Eri's default grenade is replaced by this.
- This type explodes on impact just like the default one but this one also spawns a persistent pool of fire of radius 3 blocks.
- This pool remains active for 10 seconds and damages just like the fire stream from the flame shot.

AIMING

- Aiming is recommended to be done via mouse (It can be implemented using keyboard keys).
- The angle at which the player can aim is determined by the facing direction. The minimum angle is 0 degrees i.e. straight horizontally and the max is 90 degrees i.e. straight vertically.
- Any angle in between 0 and 90 should also work.



COLLECTIBLES

FOOD

- Turkey or fruit.
- Has a small chance of dropping when an enemy is killed.
- Serves the purpose of replenishing the player's saturation stat.
- Turkey replenishes 3 units and fruit replenishes 2 units.



POW PRISONER

- A prisoner of the Morden's army waiting for you to free him.
- When in close proximity a press of a key releases it and as a thank you it drops a supply crate for you and vanishes.

SUPPLY CRATE

- The supply crate replenishes the weapons or ammo of the character.
- It will always contain a weapon and some ammo.
- Player can choose to pickup that weapon and replace it with the one already yielded.
- In case of not picking it up, the crate replenishes the existing ammo of the player.
- The Crate also contains 3-5 Hand grenades or 1-3 Fire bomb grenades.
- For Eri, the hand grenades convert to fire bomb grenades.
- There is a 90% chance that the weapon in the crate is either an HMG, a flame shot or a rocket launcher with ammo. Or a 10% chance that it is a laser gun.



FANNUM TAX

The Fat player's arrival introduces its own gravitational field. This means that when bullets get into its range, even when they were not hitting it initially in its non-fat state, now they bend their trajectory onto the fat player.



GAME MODES

Your game must support two distinct play modes, each demanding a completely different approach to level management and memory architecture. The game's main menu must allow the player to choose between them before play begins.

THE GAUNTLET

a.k.a Survival Mode

The mode consists of 3 Levels and a Boss Level. All the levels must have Vertical and Horizontal Scrolling implemented in it.

Each level is to be made of **2D Voxels** where the **width** must be between 150-250 blocks per level and the **height** must be between 20-60 blocks per level.

Do note that all the enemies should have predefined positions backed into the respective levels.

The Anatomy of the levels:

All the levels must consist of 3 specific biomes

- **Aerial:** The altitudes are high in these regions consisting of mountainous terrain with jagged peaks
- **Plains:** Central altitudes where we see plainer terrain with small hills.
- **Aquatic:** Lowest Altitudes utilizing a fixed sea level which must be vast enough to have ships sailing and must be deep enough for underwater fights.

Note: Difficulty should increase level by level (*not anatomy by anatomy*)



LEVEL 1

- Infantry must spawn 2 batches of each enemy evenly distributed in all three biomes
- Aerial and Alien must spawn 1 batches of each enemy in the aerial biomes only.
- Undead must spawn 2 batches of each enemy in plains biomes only.
- Must spawn
 - 2 batches of zombies in the Aquatic Biome.
 - 2 batches of flying Tara during the level.
 - 1 POW prisoner per biome.
 - 1 Enemy sub in Aquatic Biome



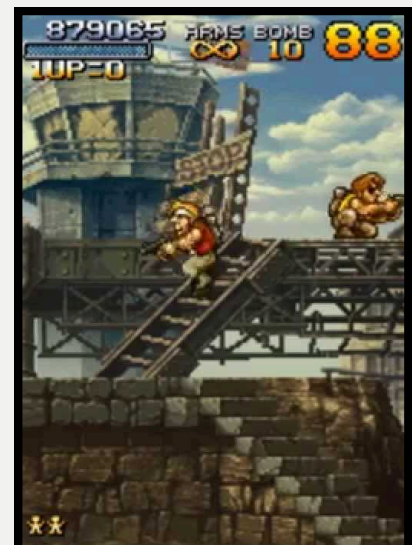
LEVEL 2



- Infantry must spawn 3 batches of each enemy in the whole distributed evenly in all three biomes
- Aerial and Alien must spawn 2 batches of each enemy in the aerial biomes only.
- Undead must spawn 3 batches of each enemy in plains biomes only.
- Must spawn:
 - 3 batches of zombies in the aquatic biome.
 - 3 batches of flying Tara during the level.
 - 2 M-15A Bradley placed in different locations.
 - 2 POW prisoners in the plains and 1 each in the remaining biomes.
 - 1 enemy sub in the Aquatic Biome

LEVEL 3

- Infantry must spawn 3 batches of each enemy in the whole distributed evenly in all three biomes
- Aerial and Alien must spawn 3 batches of each enemy in the aerial biomes only.
- Undead must spawn 4 batches of each enemy in plains biomes only.
- Must spawn
 - 4 batches of zombies in the aquatic biome.
 - 3 batches of flying Tara during the level.
 - 2 M-15A Bradley placed in different locations.
 - 2 POW prisoners per biome.
 - 2 aquatic subs in the Aquatic Biome



BOSS LEVEL

The Boss level is just like the 3 survival level with horizontal and vertical scrolling but instead of the normal enemies, it will spawn bosses.

PHASE 1

The player starts in the plains biome and faces the ground boss. When the ground boss' HP drops to 50% it retreats to outside the screen and disappears.

The player has to move ahead to the next biome in that level i.e. the aerial biome.

PHASE 2

In the Aerial Biome, the player encounters the aerial boss.

When the aerial boss' HP drops to 50%, it retreats to outside the screen and disappears.

The player has to move ahead to the next biome in that level i.e. the aquatic biome.

PHASE 3

In the Aquatic Biome, the player encounters the aquatic boss.

A slug mariner is parked on the coast to be used in the under water battle

When the aquatic boss' HP drops to 50%, it retreats to outside the screen and disappears.

The player has to move ahead to the next biome in that level i.e. the blended one with all three biomes

PHASE 4

In this phase, the player moves to a blend of two biomes. A section of the aquatic biome and a section of the plains biome.

All three retreat bosses reappear, collide violently with an explosion and fuse together to form the ultimate boss which can fly like the aerial boss, battle on the ground like the ground boss and swim like the aquatic boss. It's all three bosses at the same time. Whenever each boss's ability is used they must be done polymorphically.

When it's on the ground it uses the ground boss sprite. When in the water it uses the aquatic boss sprite and when flying it uses the aerial boss sprite.

It transitions from one state to the next in a cycle after every 10-15 seconds.

It has separate healths of all the fused bosses and when in one state that health depletes the boss can't transition to that state.

To overwhelm the player. The bosses (phases 1-4) spawn minions of the same type as the boss. From that type it chooses an enemy and spawns minions in the form of that enemy in batches of 2-4.

After every batch kill a crate spawns to replenish the ammo of the player.



BOSS ENEMIES

IRON NOKANA (Ground Boss)

- A missile launcher turret on the top and a hidden flamethrower on its bottom
- The flame thrower behaves like the flame shot and occasionally throws fire bomb grenades.
- Has 30 HP
- Spawns in a strictly flat region



HAIRBUSTER RIBERTS (Aerial Boss)



- Is equipped with several forms of armaments, such as rockets and fire bomb grenades.
- Has 30 HP.
- Spawns in high altitude regions

SEA SATAN (Aquatic Boss)

- The Sea Satan is a huge submarine.
- It can attack using its guns that fire like an HMG with one fitted on either side.
- It also fires rockets towards the players.
- Has 30 HP



THE INFINITE WORLD

a.k.a Campaign Mode

This is a single Level. It generates **infinite terrain** towards the left and towards the right of the player who could theoretically go 1 million blocks in one direction. For this purpose, you would have to leverage procedural generation via fractal noise.

Fractal Noise:

Fractal noise is n number of Perlin Noise layers stacked on top of each other for more diversity and detail.

(Links and reference documents have been provided to you for both these concepts to help you understand them as you would have to implement the noise from scratch)

The implementation of this Fractal Noise will be responsible to generate the height map which will then be mapped to the biomes:

- Aerial
- Plains
- Aquatic

This single level will be considered complete **until 5 of each type** of enemies have been killed. 3 of each enemy vehicles must also be destroyed.

Injecting OOP into the veins of Perlin Noise

When generating the World/level in campaign mode the player is given an option to choose a Perlin Noise profile to choose from that will set the inputs of the noise function resulting in terrain features.

- **Amplified:** The peaks and the ocean are extra amplified while the plains remain the same.
- **Flat:** Opposite of Amplified. No dramatic peaks and deep oceans.
- **Normal:** The normal profile where the depth of the ocean and the altitude of the peaks is normal as in uniformly distributed.

You must implement a level creator profile factory to generate these profiles and then use pointers of an abstract class to polymorphically choose the level profile at runtime.

Note: Enemy Spawning will be done dynamically at runtime (No fixed positions). To complete the single Campaign Level, you must kill 5 of each of the enemies. The player must also destroy 3 of each enemy vehicle.

RULES FOR BOTH MODES

- Blocks / Voxels in both modes must be destructible via blasts if they fall into the blast radius.
- If the craters are created ≤ 3 blocks wide and ≥ 1 block deep, then this crater **must fill up** with water **if** exposed to rain for a certain amount of time.
- Rain is an environment feature which can happen at a random time for a random interval.
- The bottom-most layer of the levels must be a special block that must be indestructible under any circumstances.
- You must spawn each of the player vehicles in both modes in each level at appropriate locations so that they can be of the most help.



THE ART OF SELF PLAYING

a.k.a AI becomes the gamer

NeuroEvolution of Augmented Topologies (NEAT) is an Evolutionary Optimization Algorithm presented by Kenneth O. Stanley and Risto Miikkulainen in 2002. The original research paper is attached with the Project file to understand the complete working of the algorithm.

You are required to add a “Self Playing” mode which is based on NEAT that lets the players automatically play the game, err; adjust and evolve both the weight and topology of the network over generations so that they produce a player that performs better than the pre generations. This evolution should happen indefinitely, making the next generation better and better automatically; till the self playing mode is exited.

It should keep a record of the global best player along with its weights and topology. This global best player should be the best run across every time the self playing mode is ever played. The keyboard key 'B' should be preserved, and press it to activate that best player alongside the current running generation *[highlighted in whatever way you deem the best]* for comparison and visual benchmarking.



Scoring

BASIC ENEMY KILLS

Class	Enemy	Score
	Rebel Soldier	50
	Shielded Soldier	75
	Bazooka Soldier	100
	Grenade Soldier	100
Aerial	Paratroopers	+25 from base infantry enemy
Undead	Mummy	150
	Zombie	100
Alien	Martian	200

BOSS KILLS

Encounter	Score
Normal Bosses (Phases 1-3)	500
Ultimate Boss (Phase 4)	1500

BONUSES ON KILLS

Feat	Description	Score
Melee Kill	Kill an enemy with a knife in close range	+50
Aerial Kill	Killing an enemy mid jump	+100
Multi-Kill	Killing 2 enemies with a grenade or a rocket	+200
Massacre	Killing 3 or more enemies with a grenade or a rocket	+300 base + n x 50 for kill more than 3

LEVEL CLEAR SCORES

Feat	Description	Score
Survival mode clear	Kill all the enemies and reach the right boundary	1000
Campaign mode clear	Reach the kill quota to start the boss level	3000
Flawless victory	Killing 2 enemies with a grenade or a rocket	5000

Special Game Features

DEVELOPER MODE

You must implement a “Developer Mode” cheat code (*your choice*) into the game that lets the player be immortal, and lets it have access to every weapon (a weapon each to every character) so that during the demo we can test and verify everything implemented without the student dying repeatedly. (Lesson learned from last year).

FORKS OF FATE

The ‘R’ key on Keyboard when pressed should enable a rewind of 10 seconds, showing a frame by frame reversal of the player and its environment including the enemies. Store the timeline in a git-like sequence where an action done if in the past, bursts out another branch of the timeline from that point.

Wherever the ‘R’ button is pressed a node (checkpoints) should form branching out alternate timelines; that should be stored using appropriate file handling techniques. You should implement a way to shift between timelines from a specific node across a run of the game.

THE REFLEX FINGERPRINT

As you play the game, the game learns your exact style of playing. The longer you play the better the game gets at imitating you. Assign a specific key which bursts out a reflex fingerprint mode during playing. This will enable the game to autopilot for 15 seconds based on your exact playing styles and imitate your patterns of attack and defense. The parameters should be at least but not limited to FingerPrint file; can add more for accurate representation.



THE FINGERPRINT FILE AND GHOST

A downloadable Player Style Profile should be implemented. It should have the exact style of the players, the parameters and their weights. The file should be uploadable to another instance of a game on a separate device and a ghost player (fainted in colours and transparency) following the exact playing style as encoded in the uploaded fingerprint file; should spawn alongside the main player. The ghost should play automatically based on the said style, varying from fingerprint file to file. The ghost inflicts 80% less damage and has 300% more health than a regular player. We have attached the format in 'parameters.csv'. This has to be the exact format so that one import can be an exported file of another

GRUDGEFUL ENEMIES

The enemies that survive your attacks and you move past them (your screen frame does move past them) do not get killed. They have a photographic memory and return backwards with a better power enhanced version. They hold a grudge against you and attack you first finding their way through obstacles.

MEMORY OF COWARDICE

The world remembers what the player leaves unfinished. Enemies that are ignored instead of killed gather in a hidden reserve and may eventually return together in a sudden, overwhelming counterattack.



ECHO CHAMBERS

When the player is submerged into the water in aquatic biomes you must use muffled sounds for sound effects and background music. The catch is that you can't use prepared muffled sounds. You must use Q4 of assignment 2 and decompose the sounds into discrete frequencies and delete the frequencies at the higher end of the spectrum. This decomposition must happen at runtime.

THE RAGEBAIT

It tracks not your actual score but your score expectation; what you historically believed a run was worth at its midpoint based on how you played the first half. When a run is going well by your own historical standards, it starts introducing low-value enemies and removing high-value configurations, so the final score lands slightly below your expectation. Never catastrophically. Always just enough disappointment to motivate one more run. It has learned the exact gap between your expected score and your actual score that keeps you playing. It maintains that gap indefinitely.

THE DEATH COMPOSITOR

Every death you've ever had is stored as a fingerprint; your health, position, active weapon, enemy configuration, time since last checkpoint. It builds a composition of your most frequent death fingerprints. When your current run state closely resembles a historical death fingerprint, the game starts gently steering the environment toward completing that pattern; not spawning the specific enemy that killed you before, but tilting spawn weights toward it, shifting enemy aggression up slightly, placing a wall just where you historically needed open space. It doesn't kill you. It recreates the conditions and lets you kill yourself. The death feels like yours. It was assembled from a library of your previous ones.

THE CHAINS OF DEATH

You are chained into a series of predictions. A prediction of states.

It maintains a small set of states more like abstract behavioral states like pressing, consolidating, retreating, burning resources, stalling (the more the better). Each state is defined by a cluster of observable signals: movement speed, grenade frequency, weapon swap rate, distance maintained from enemies, time spent near cover versus in open space (extra points will be awarded for more parameters). You are always in exactly one state. The state you're in right now is inferred from the last 4 seconds of your inputs.

The chain is the transition table. From pressing, for example, you historically move to burning resources 40% of the time, stay in pressing 35%, and drop to consolidating 25%. From consolidating, you almost never go directly to retreating, you pass through stalling first, almost always.

The chain predicts your next state, not your next action. It doesn't know you're about to throw a grenade. It knows you're about to enter burning resources, which means a grenade or a rocket or a slug slam is coming in the next 6 seconds. That's enough information it needs to paralyze you.

The enemy spawner factory reads the chain's current prediction. If you're about to enter burning resources, it holds the next heavy enemy spawn for 7 seconds; lets you bait into waste the resource on nothing, then sends the heavy infantry. If you're in stalling, it accelerates spawns because stalling players historically respond to pressure by either snapping into pressing or collapsing into retreating, and both are exploitable transitions.

The chain has a second use. Some states are traps. For example the chain figures out that Consolidating followed by pressing is your highest-performance sequence you reset, then commit, and you play well. The game identifies when you're entering that sequence and introduces a single disruption; one offscreen bullet, one mis-timed spawn ; calibrated to knock you out of consolidation before you complete it. You never get the clean reset. You always go into pressing slightly unready. The disruption is when you least expect it knocking you out at your lowest moments.

The boss uses the chain differently. It watches your state transitions for the first 20 seconds of the fight, establishes your current transition probabilities under pressure, and identifies the state you spend the least time in. That state is your gap like it understands the behavioral configuration you avoid or move through too quickly. The boss then engineers conditions that force you into that state and hold you there. If you almost never stall, the boss creates a situation where the correct response is to wait, and waits with you, because it knows waiting is where you break.

The chain remembers across sessions. Your transition probabilities from last week are in there, weighted less than today's but still present. If you spent a month playing aggressively and just came back after a break, the chain has both versions of you- the aggressive one and whatever you are now; the weights are decided between them by recency. The first 10 minutes of a returning session are the most dangerous. The chain is running on old you while you're still figuring out what you currently are.

URDU ASSEMBLY INJECTION

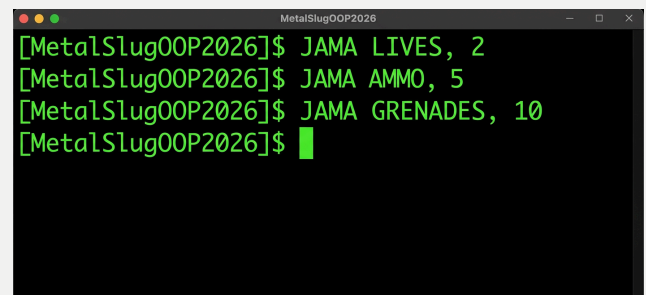
When the developer mode key is pressed thrice in 2 seconds a terminal window spawns letting you input commands in Urdu assembly from Assignment 3.

Using the syntax and reserved words from Urdu Assembly you can add, or subtract from stats or variables like health, score, ammo etc.

For Example.

JAMA LIVES, 2
JAMA AMMO, 5
JAMA GRENADES, 10

This dynamically modifies the behaviour of the game at runtime. You are able to Run Urdu Assembly in the terminal directly. Think of it as similar to how Python's Interactive mode runs. This is direct injection into runtime behaviour of the game.



```
[MetalSlugOOP2026]$ JAMA LIVES, 2
[MetalSlugOOP2026]$ JAMA AMMO, 5
[MetalSlugOOP2026]$ JAMA GRENADES, 10
[MetalSlugOOP2026]$ █
```

THE VOICE OF THE BENE GESSERIT

Think of this as a Startup Engine. Use windows.h to link up to the device's microphone and as the game's instance starts; the microphone starts listening for the magic phrase "Valar Morghulis" (the exact audio is provided amongst the project file). The provided audio must be decomposed using fast fourier transformations (*Q4 mentioned rawrrr*) and stored into matrices. The audio sampling coming from the microphone must be decomposed in similar fashion. You need to account for the environmental damages to the listened audio as that would produce a blurred version of the original audio; so exact matching must not be done but rather cosine similarity be applied between both versions and anything that matches 40% of the original sound must trigger the game to start. A popup of "Command Detected Successfully" upon detecting the phrase.

References

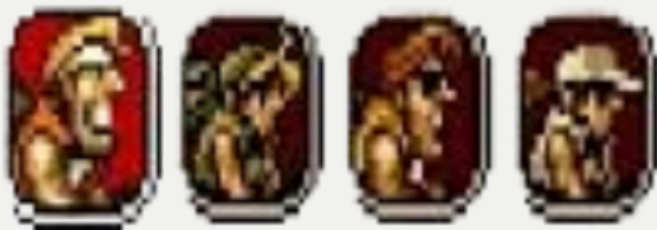
[Minecraft terrain generation in a nutshell](#)
[How Does Perlin Noise Work?](#)



Genuine Question BTW

WHO WOULD WIN?

THE METAL SLUG TEAM



THE SAME
PEOPLE BUT FAT



Best Of Luck